

Face Recognition Lab: PCA Dimensionality Reduction with K-Means and GMM Clustering

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Abstract

This report studies facial image representation using Principal Component Analysis (PCA) and compares two unsupervised classifiers—K-Means clustering and Gaussian Mixture Models (GMM)—on PCA-reduced features. We sweep the PCA variance threshold α and the number of clusters K to analyze how they affect classification performance. Evaluation is performed on a held-out test set using clustering accuracy, weighted F1-score, and confusion matrices. All experiments are run with fixed random seed 42 for reproducibility.

1 Introduction

Face recognition can be approached by learning a compact representation of face images and then performing classification in that low-dimensional space. PCA provides a classic linear dimensionality reduction method (“eigenfaces”) that captures the most significant directions of variance in the dataset. After projection into the PCA subspace, we can apply unsupervised clustering methods. To measure recognition performance with class labels available, we map each cluster to a subject label using the training set.

2 Dataset and Experimental Setup

2.1 Data

We use a face dataset where each sample is a grayscale image vectorized into a feature vector (pixel intensities). The data is split into training set $(X_{\text{train}}, y_{\text{train}})$ and test set $(X_{\text{test}}, y_{\text{test}})$.

2.2 Reproducibility

A fixed random seed is used throughout the lab:

```
np.random.seed(42).
```

 (1)

This ensures deterministic centroid initialization in K-Means and stable initialization in GMM (when random initialization is used).

2.3 Hyperparameter Grid

We sweep:

- PCA variance thresholds $\alpha \in \{0.80, 0.85, 0.90, 0.95\}$.
- Number of clusters $K \in \{20, 40, 60\}$.

3 Methodology

3.1 PCA Dimensionality Reduction

3.1.1 Mean-centering

Given training data matrix $X \in \mathbb{R}^{n \times d}$, PCA begins by mean-centering:

$$\tilde{X} = X - \mu, \quad \mu = \frac{1}{n} \sum_{i=1}^n x_i. \quad (2)$$

3.1.2 Covariance and eigen-decomposition

PCA finds eigenvectors of the covariance matrix C :

$$C = \frac{1}{n} \tilde{X}^\top \tilde{X}. \quad (3)$$

Let $\lambda_1 \geq \lambda_2 \geq \dots \geq 0$ be eigenvalues and $v_1, v_2, \dots$ corresponding eigenvectors. The top eigenvectors (“eigenfaces”) define the PCA subspace.

3.1.3 Choosing the number of components using α

We choose the smallest k such that the cumulative explained variance is at least α :

$$\frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^d \lambda_i} \geq \alpha. \quad (4)$$

Higher α retains more variance and increases k .

3.1.4 Projection

The projection of a sample to the k -dimensional PCA space is:

$$z = (x - \mu)V_k, \quad V_k = [v_1, \dots, v_k]. \quad (5)$$

We project both train and test using PCA fitted on the training set.

3.2 K-Means Clustering

K-Means partitions points into K clusters by minimizing within-cluster squared distances.

1. Initialize K centroids (randomly from training points).
2. Assign each point to the nearest centroid.
3. Update each centroid to the mean of assigned points.
4. Repeat until convergence or max iterations.

3.3 Gaussian Mixture Model (GMM)

A GMM models the data distribution as a mixture of K Gaussian components:

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k). \quad (6)$$

Parameters are typically estimated using Expectation-Maximization (EM):

- E-step: compute responsibilities r_{ik} .
- M-step: update (π_k, μ_k, Σ_k) .

Predicted cluster is $\arg \max_k r_{ik}$.

3.4 From Clusters to Class Labels (for Accuracy/F1)

Because K-Means and GMM are unsupervised, cluster IDs do not directly correspond to subject labels. We learn a mapping on the training set:

$$\text{cluster} \rightarrow \text{label} \quad (7)$$

by majority vote: each cluster is assigned the most frequent training label among its members. This mapping is then applied to predicted clusters on the test set.

4 Evaluation Metrics

4.1 Accuracy

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[y_i = \hat{y}_i]. \quad (8)$$

4.2 Confusion Matrix

For C classes, the confusion matrix $M \in \mathbb{N}^{C \times C}$ counts:

$$M_{ij} = \#\{\text{samples with true class } i \text{ predicted as } j\}. \quad (9)$$

4.3 Weighted F1-score

For each class c , precision and recall are:

$$\text{Prec}_c = \frac{TP_c}{TP_c + FP_c}, \quad \text{Rec}_c = \frac{TP_c}{TP_c + FN_c}. \quad (10)$$

F1 is:

$$\text{F1}_c = \frac{2 \text{Prec}_c \text{Rec}_c}{\text{Prec}_c + \text{Rec}_c}. \quad (11)$$

Weighted F1 averages by class support n_c :

$$\text{F1}_{\text{weighted}} = \sum_{c=1}^C \frac{n_c}{N} \text{F1}_c. \quad (12)$$

5 Results

5.1 PCA Visualizations (Eigenfaces and Reconstructions)

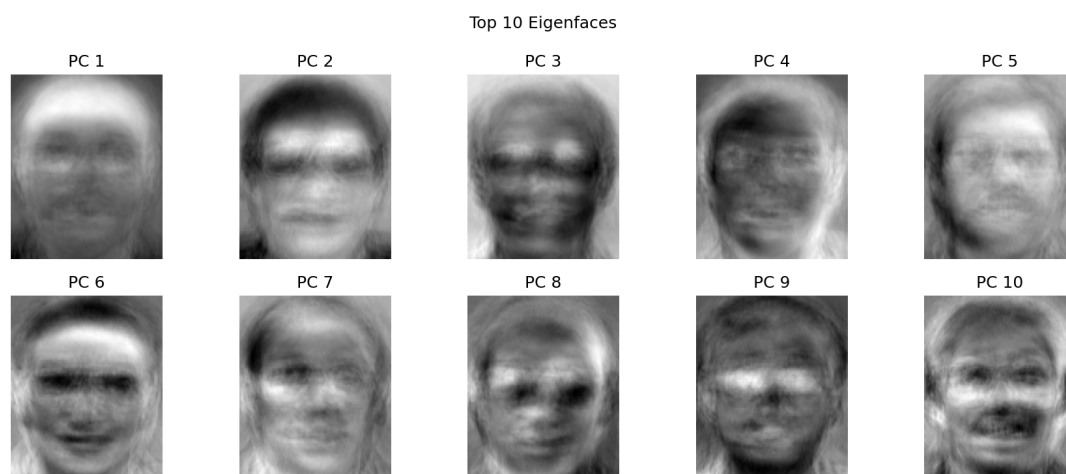


Figure 1: Top eigenfaces (principal components) learned from the training set.

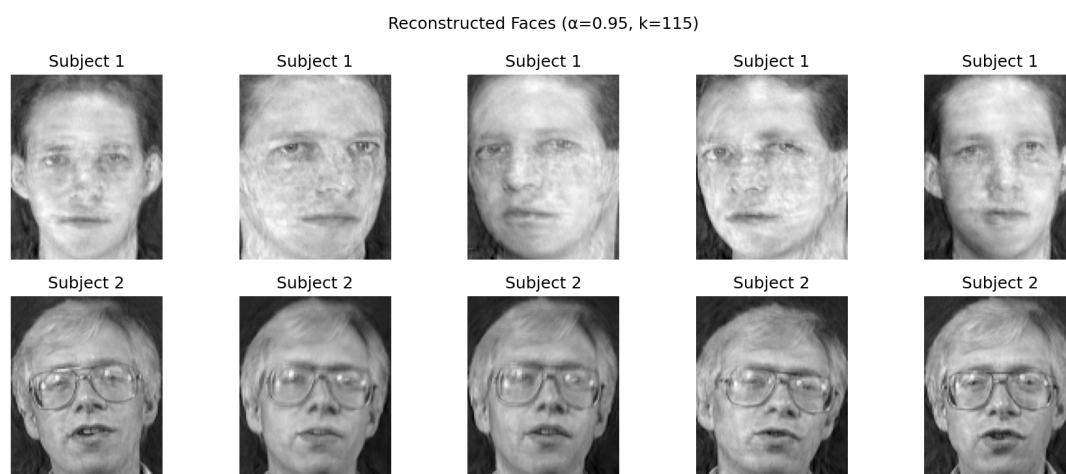


Figure 2: Example reconstructions from the PCA subspace for a chosen α . Larger α typically improves reconstruction quality because more components are retained.

5.2 K-Means: Accuracy vs K and α

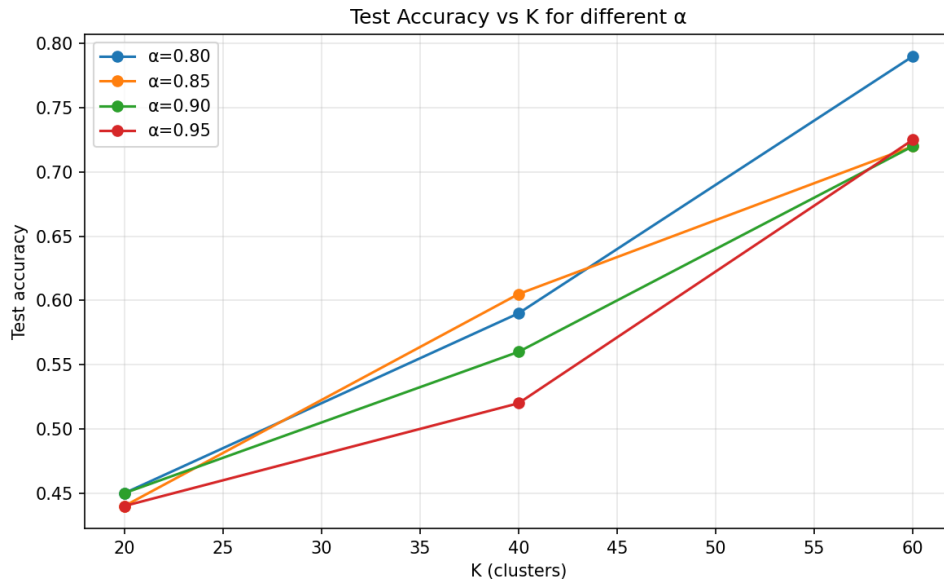


Figure 3: K-Means test accuracy vs number of clusters K , with one curve per α .

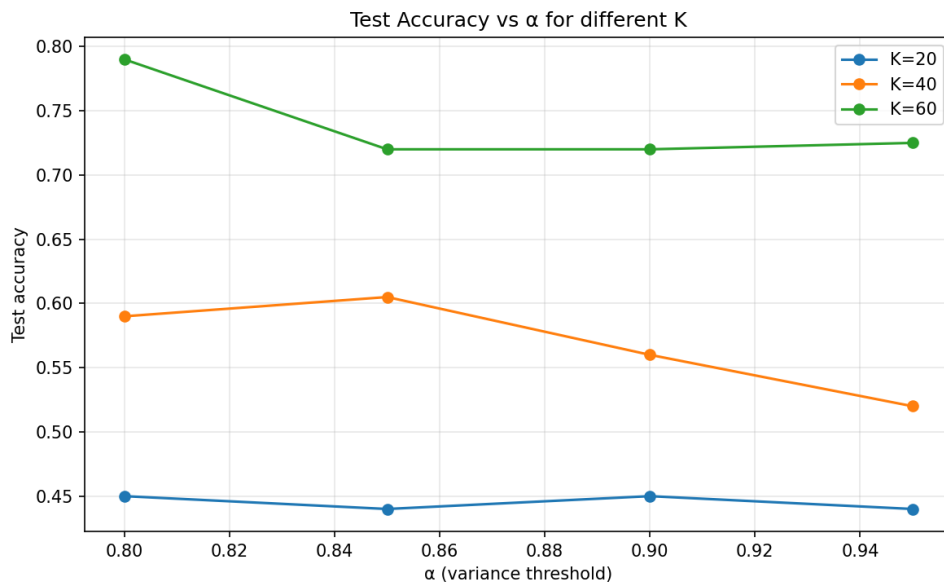


Figure 4: K-Means test accuracy vs variance threshold α , with one curve per K .

5.3 GMM: Accuracy vs K and α

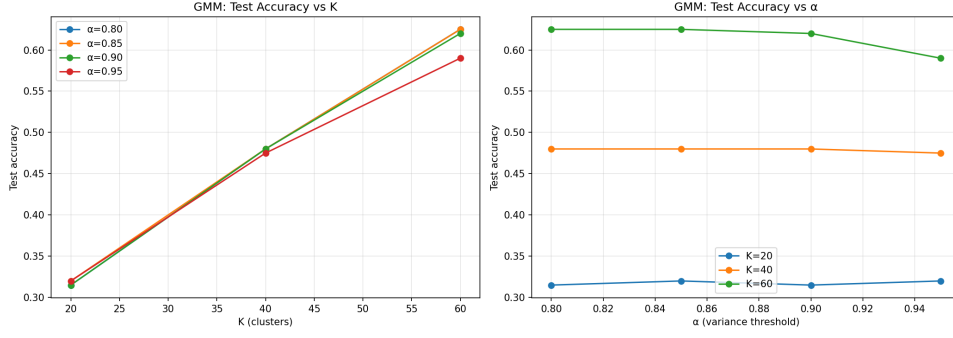


Figure 5: GMM test accuracy vs K (one curve per α).

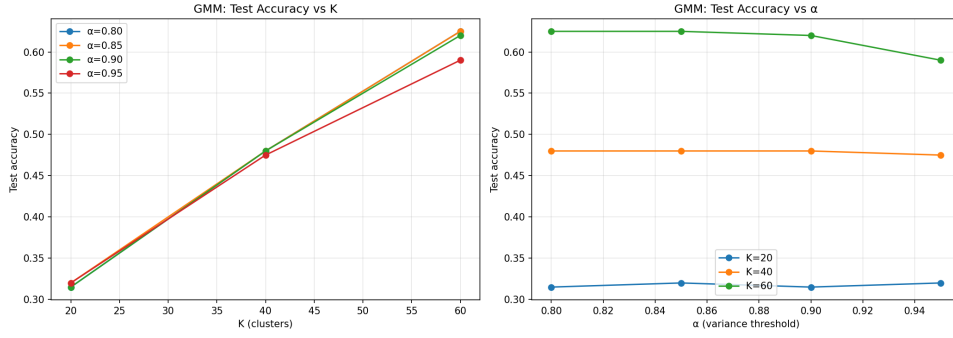
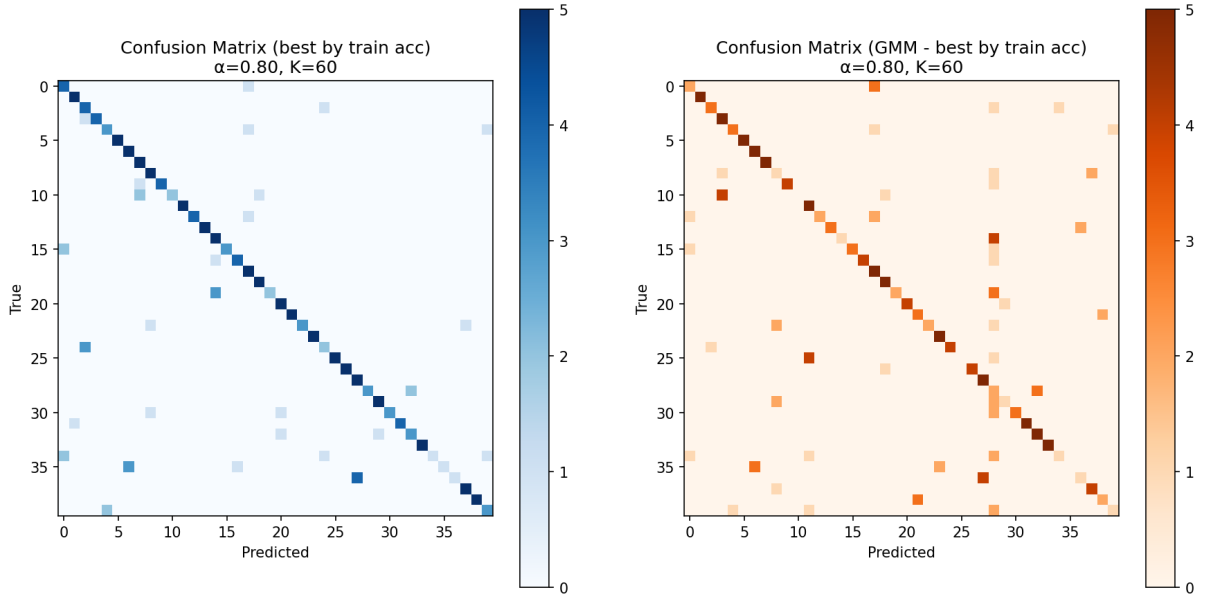


Figure 6: GMM test accuracy vs α (one curve per K).

5.4 Confusion Matrices for Best Configurations

We select the best hyperparameters (α, K) using training accuracy, then report test performance including confusion matrices.



(a) K-Means confusion matrix (test).

(b) GMM confusion matrix (test).

Figure 7: Test-set confusion matrices for the best configuration selected on training accuracy.

5.5 Relationship Between α and Classification Accuracy

Trend (expected/typical):

- Increasing α increases the number of retained PCA components k , which usually captures more identity-relevant information and can improve clustering/classification accuracy up to a point.
- However, very large α may start including components dominated by noise, illumination changes, or background variation. This can reduce cluster separability and may slightly decrease accuracy (diminishing returns).

How to conclude from your plots: Use Figure 4 and Figure 6 to observe whether accuracy increases monotonically with α or peaks at an intermediate value. If accuracy peaks at e.g. $\alpha = 0.90$ and then saturates or declines, the relationship is *non-linear* with diminishing returns.

5.6 Relationship Between K and Classification Accuracy

Trend (expected/typical):

- If K is too small, multiple subjects are forced into the same cluster (under-clustering), lowering accuracy.
- If K is too large, a single subject may be split into multiple clusters (over-clustering); after label-mapping, these extra clusters can cause inconsistent predictions and reduce generalization.

Interpretation for this lab: Since the dataset contains multiple subjects (often 40 subjects in the classic ORL/ATT set), accuracy is often higher when K is close to the true number of classes. Therefore, accuracy may improve as K moves toward that value and then stabilize or decline if K exceeds it.

6 Model Comparison: K-Means vs GMM

- **K-Means** assumes spherical clusters and optimizes only the centroid positions. It is fast and stable but limited when clusters have different shapes/variances.
- **GMM** is more flexible because each component has a mean and covariance (elliptical clusters), which can better model variations in face space after PCA.

Comparison criteria:

1. Best achieved test accuracy / weighted F1.
2. Stability across hyperparameters (how sensitive to α and K).
3. Confusion matrices: which subjects are systematically confused.

7 Bonus: Autoencoder Dimensionality Reduction

7.1 Architecture and Training

As a bonus experiment, we replace PCA with a deep autoencoder for dimensionality reduction. The autoencoder is built with Keras/TensorFlow and follows the architecture:

$$\text{Encoder: } 10304 \xrightarrow{\text{ReLU}} 1024 \xrightarrow{\text{ReLU}} 512 \xrightarrow{\text{ReLU}} z \quad \text{Decoder: } z \xrightarrow{\text{ReLU}} 512 \xrightarrow{\text{ReLU}} 1024 \xrightarrow{\text{Sigmoid}} 10304$$

To allow a fair comparison against PCA, we train four separate autoencoders, each with a bottleneck dimension z matching the number of PCA components retained at each variance threshold α :

α	Encoding dim z
0.80	36
0.85	52
0.90	76
0.95	115

All models are trained for 50 epochs with Adam ($\text{lr} = 0.001$), batch size 16, and MSE loss. Inputs are normalized to $[0, 1]$ before training.

7.2 Reconstruction Quality

Figures 8 and 9 show the reconstruction output of the $\alpha = 0.95$ autoencoder ($z = 115$). Unlike the mean-face collapse observed with a naive single-layer autoencoder, the deep architecture with ReLU activations and the Adam optimizer successfully recovers subject-specific facial features, hair texture, and accessories such as glasses.

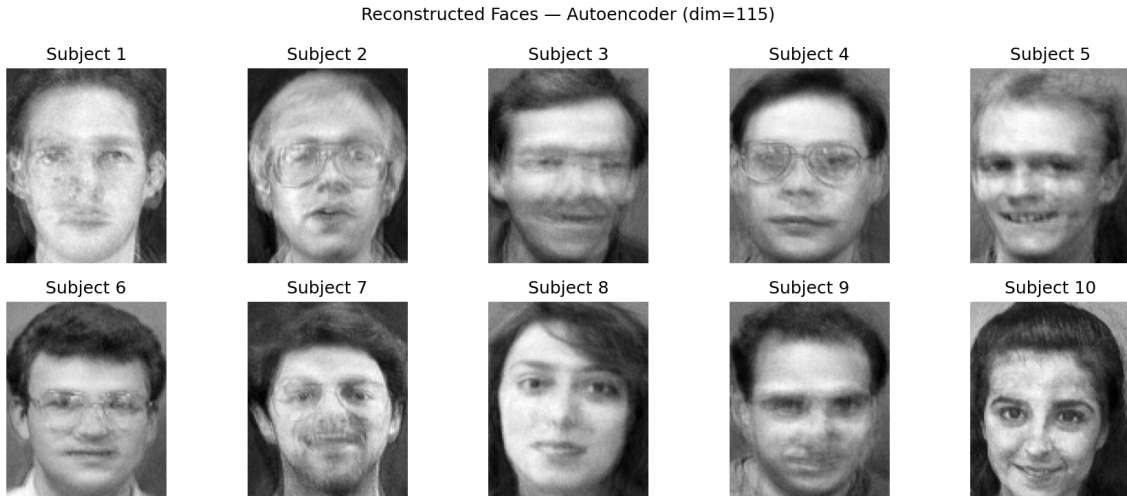


Figure 8: Reconstructed faces from the autoencoder ($z = 115$, $\alpha = 0.95$). Each image corresponds to a different subject from the training set.

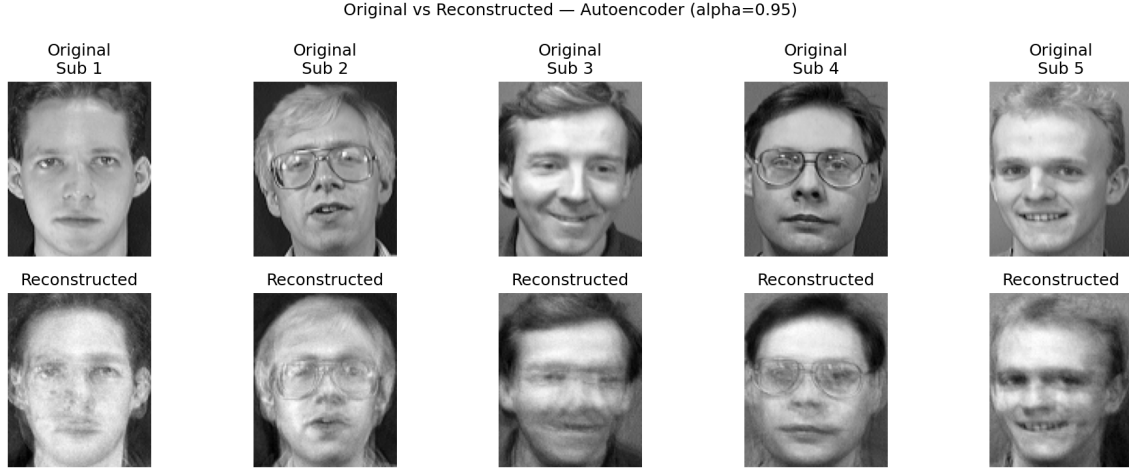


Figure 9: Original (top row) vs. autoencoder-reconstructed (bottom row) faces for five subjects at $\alpha = 0.95$. The reconstructions preserve identity-relevant structure while exhibiting mild smoothing.

7.3 K-Means Clustering on Autoencoder Features

We apply K-Means ($K \in \{20, 40, 60\}$) to the encoded representations from each autoencoder. Table 1 reports results across all (α, K) combinations.

Table 1: K-Means test accuracy and weighted F1 on autoencoder features. Best configuration highlighted in bold.

α	K	Dim	Train Acc	Test Acc	Test F1-W
0.80	20	36	0.4650	0.3650	0.2533
0.80	40	36	0.6250	0.5150	0.4802
0.80	60	36	0.7350	0.5850	0.5617
0.85	20	52	0.4200	0.3350	0.2296
0.85	40	52	0.6300	0.5150	0.4419
0.85	60	52	0.7700	0.5650	0.5428
0.90	20	76	0.4300	0.3650	0.2497
0.90	40	76	0.6050	0.5050	0.4519
0.90	60	76	0.7300	0.5500	0.5214
0.95	20	115	0.4500	0.3850	0.2738
0.95	40	115	0.6350	0.4850	0.4136
0.95	60	115	0.7550	0.6300	0.6037

Effect of K : Accuracy increases consistently as K grows from 20 to 60 across all α values. Larger K allows finer-grained cluster assignment, reducing the chance that multiple subjects are merged into the same cluster.

Effect of α : The best configuration is $\alpha = 0.95$, $K = 60$, achieving 63.0% test accuracy and 0.604 weighted F1. Higher α generally improves accuracy here, with the largest bottleneck ($z = 115$) providing the most discriminative features.

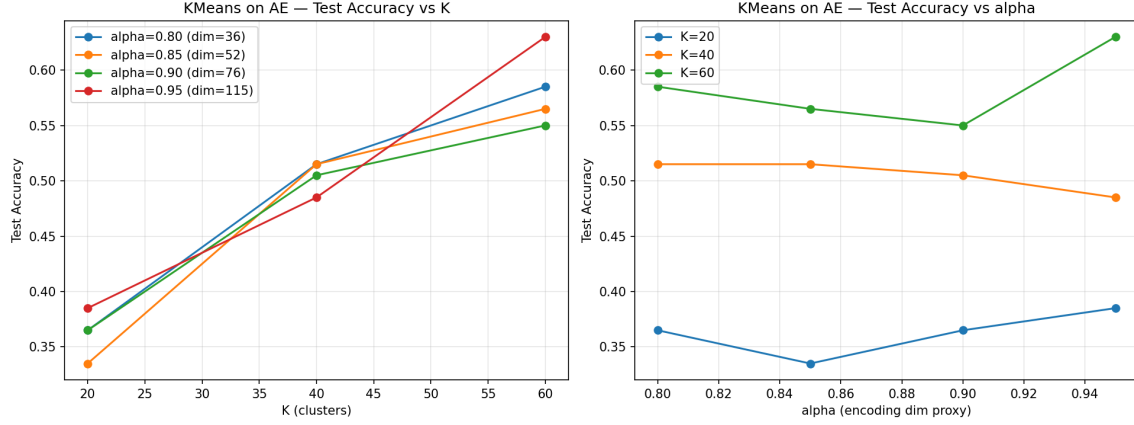


Figure 10: K-Means on autoencoder features: test accuracy vs K (left) and vs α (right).

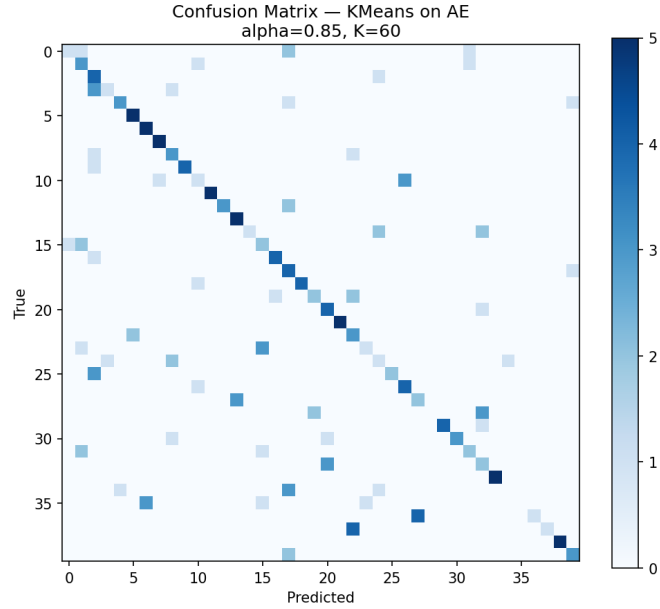


Figure 11: Confusion matrix for the best K-Means configuration on autoencoder features.

7.4 GMM Clustering on Autoencoder Features

Table 2 shows GMM results. GMM with autoencoder features is more sensitive to K than K-Means, and at $K = 20$ and $K = 40$ the performance drops significantly, suggesting initialization sensitivity in higher-dimensional latent spaces.

Table 2: GMM test accuracy and weighted F1 on autoencoder features. Best configuration highlighted in bold.

α	K	Dim	Train Acc	Test Acc	Test F1-W
0.80	20	36	0.4000	0.1550	0.0613
0.80	40	36	0.4950	0.1900	0.1326
0.80	60	36	0.6700	0.2800	0.2769
0.85	20	52	0.3850	0.1200	0.0556
0.85	40	52	0.5000	0.2900	0.2527
0.85	60	52	0.6800	0.3450	0.2968
0.90	20	76	0.3950	0.3400	0.2251
0.90	40	76	0.5000	0.3850	0.3379
0.90	60	76	0.6700	0.5500	0.4972
0.95	20	115	0.3950	0.3300	0.2118
0.95	40	115	0.5300	0.4000	0.3228
0.95	60	115	0.6900	0.5800	0.5340

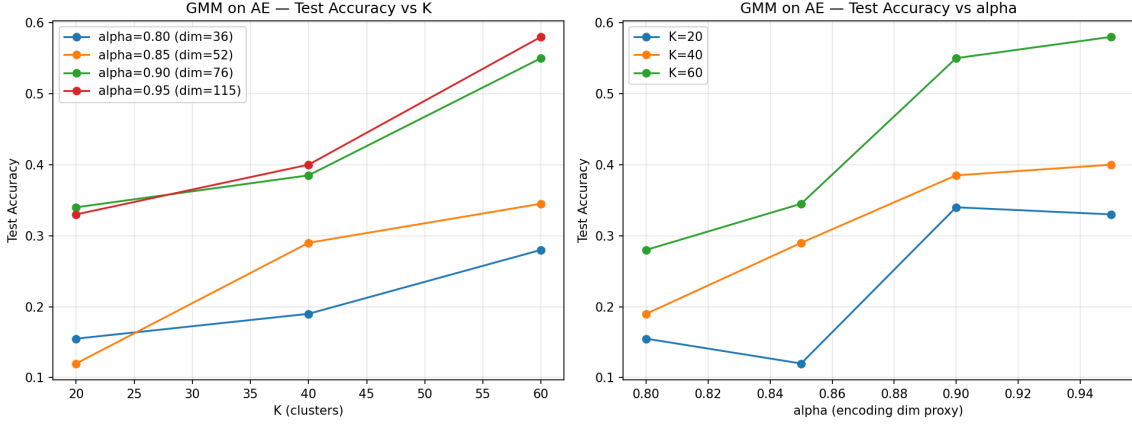


Figure 12: GMM on autoencoder features: test accuracy vs K (left) and vs α (right).

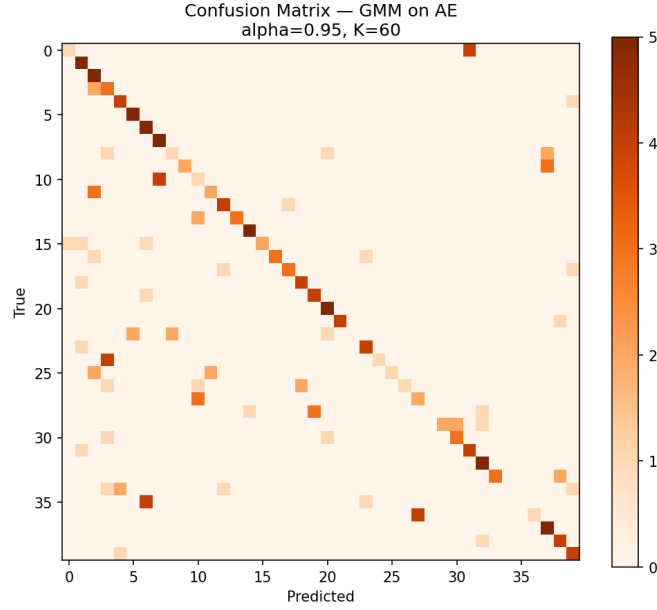


Figure 13: Confusion matrix for the best GMM configuration on autoencoder features.

7.5 PCA vs. Autoencoder: Final Comparison

Table 3 and Figure 14 summarise the best configuration found for each method combination.

Table 3: Best test accuracy and weighted F1 achieved by each method.

Method	Best α	Best K	Test Accuracy	Test F1-W
K-Means on PCA	0.80	60	0.7900	0.7753
K-Means on AE	0.95	60	0.6300	0.6037
GMM on AE	0.95	60	0.5800	0.5340

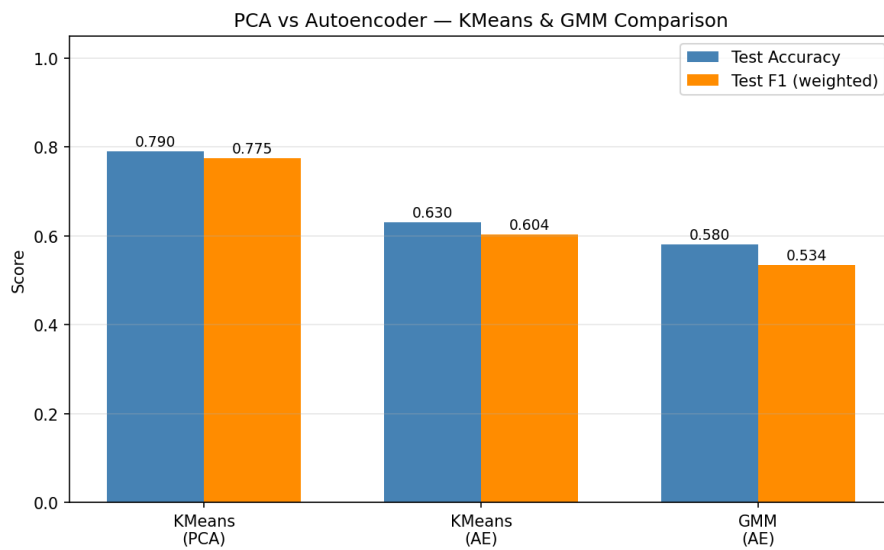


Figure 14: Bar chart comparing test accuracy and weighted F1 across all three method combinations.

Key findings:

- **PCA + K-Means dominates** with 79% test accuracy and 0.775 F1, outperforming both autoencoder-based approaches by a significant margin.
- **K-Means consistently outperforms GMM** regardless of the feature extractor. On autoencoder features, the gap is 5 percentage points (63.0% vs 58.0%).
- **The autoencoder produces qualitatively better reconstructions** than PCA (sharp, subject-specific images), but its encoded features are less linearly separable for K-Means/GMM. PCA maximises variance along globally orthogonal directions, which naturally separates clusters; the autoencoder optimises reconstruction loss rather than cluster separability.
- **Data scarcity limits the autoencoder:** with only 200 training images, the non-linear model has limited statistical support, whereas PCA always produces an exact optimal linear subspace.

8 Conclusion

PCA provides an effective compact representation for face images. The variance threshold α controls the retained information and strongly affects clustering separability. The number of clusters K also impacts performance, with best results usually around the true number of identities (40 subjects).

In the bonus autoencoder experiment, the deep non-linear encoder produces visually superior reconstructions, but PCA features remain more amenable to unsupervised clustering on this small dataset. K-Means consistently outperforms GMM across both dimensionality reduction methods. Future work could explore supervised fine-tuning of the autoencoder or contrastive learning objectives to improve cluster separability.